TEXORA · LECTURE SHEET

Physical and Mechanical Finishing

Heat, moisture, pressure, tension and cutting edges — how machines alone flatten, lustre, raise, shear, shrink-proof and heat-set a fabric without any chemical reaction.

WP-4201 · TEXTILE FINISHING · TOPIC 2 · SPECIALIZATION — WET PROCESSING

1. Principle of mechanical finishing

Mechanical (physical) finishes change a fabric by **rearranging its structure**: yarns are flattened or rounded, fibres are lifted or cut, loops and threads are compacted lengthwise, and thermoplastic fibres are re-set in a new shape. The tools are temperature, moisture content, pressure, tension/overfeed and time — the same five variables, in different combinations, produce every finish in this sheet. No new substance is added (apart from water or steam), so the effect depends on how permanently the new structure is locked: a calender lustre on cotton relaxes away in the first wash, whereas compressive shrinkage and polyester heat setting survive the life of the garment.

2. Decatising

Decatising (decatizing) is the final steam-and-pressure treatment of wool and wool-blend woven fabrics, giving a smooth, lustrous, dimensionally settled cloth with the characteristic worsted handle. The fabric is wound, interleaved with a cotton or synthetic *wrapper cloth*, onto a perforated cylinder. In **wet decatising** steam is blown through the roll (typically several minutes, followed by vacuum cooling and unwinding); in **dry (blow) decatising** the roll is steamed in an autoclave-type machine. Hot moisture plasticises the wool keratin, internal stresses from weaving and earlier finishing relax, and the fabric sets flat as it cools under tension. Effects: permanent press-like set for wool, removal of watermarks and shine marks, improved lustre and handle, and better dimensional stability.

3. Calendering

A calender passes fabric between two or more heavy rollers (bowls) under high pressure, with heat when required. The nip flattens the yarns, closes the interstices, smooths the surface and develops lustre by presenting a more uniform reflecting plane. A typical bowl stack combines a heated metal bowl against cotton, paper or polymer bowls (Figure 1).

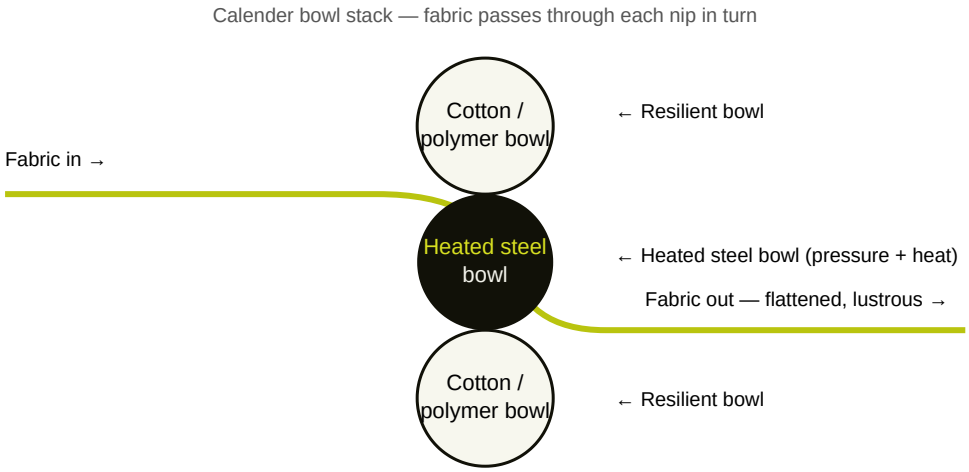


Figure 1 — Three-bowl calender stack (schematic). Yarn flattening at the nips produces smoothness and lustre.

CALENDER TYPE	CONSTRUCTION / ACTION	EFFECT AND USE
Swizzing (ordinary)	Fabric passes once through the nips of a multi-bowl stack	General smoothing, closing and lustre for cotton and blends
Chasing	Several layers of fabric pass through the nips together	Soft, linen-like, thready lustre; layers impress texture on each other
Friction calendering	Heated steel bowl rotates faster than the fabric feed (surface speed difference)	High, glazed lustre (chintz); friction polishes the surface
Embossing	Engraved steel bowl against a soft bowl, heated	Permanent raised pattern pressed into the fabric (durable on thermoplastics and resin-treated cotton)
Schreiner calendering	Steel bowl engraved with very fine diagonal lines (hundreds per inch)	Silk-like soft lustre with fine line texture on cotton — the classic Schreiner finish

Calender effects on untreated cellulosics are **temporary**: washing swells the yarns back to roundness and the lustre falls. On resin-finished cotton, and on heat-set synthetics, embossed and Schreiner effects are far more durable because the flattened state is chemically or thermally locked.

4. Sanforizing — compressive shrinkage

Woven cotton leaves the loom and the wet processes under warp tension. In washing, the warp yarns relax and crimp more, so the fabric shrinks lengthwise — often 5–10% in an unshrunk cloth. **Sanforizing** pre-shrinks the fabric mechanically, under control, so that the *residual* shrinkage left for the consumer is small — the Sanforized standard is commonly ≤1%.

Principle and machine action

The damp fabric is fed onto an endless thick **rubber belt** that is stretched as it passes over a feed roller and then allowed to contract as it goes round a steam-heated cylinder (Figure 2). While the belt surface contracts, it carries the fabric with it and compresses it lengthwise: warp threads are pushed closer together, warp crimp increases, and the length the fabric would lose in its first wash is taken out in the machine. A blanket/cloth and pressure roller keep the fabric in intimate contact with the belt through the shrinkage zone; the fabric is then dried on a cylinder (Palmer) unit without tension.

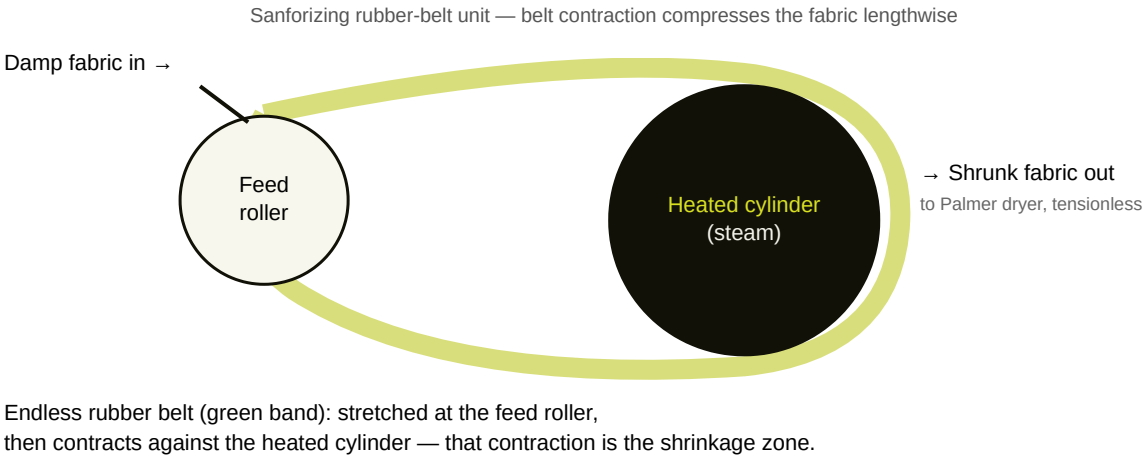


Figure 2 — Rubber-belt compressive shrinkage (Sanforizing) unit, schematic. Belt surface contraction in the shrinkage zone sets the percentage shrinkage given.

The percentage is set by how much the belt is stretched before the nip — belt thickness, feed-roller setting and pressure control the contraction, typically adjustable over about 0–12%. The operator proves the result with a wash test on a marked specimen, not from the machine dial alone.

WORKED EXAMPLE 1 — LENGTH DELIVERED AFTER SANFORIZING

A 50 m length of cotton fabric is given 6% compressive shrinkage.
Length removed = 6% × 50 m = 3.0 m. Delivered length = 50 − 3.0 = **47.0 m**.
The mill must therefore start with more grey length for every finished metre ordered — sanforizing shrinkage is a real yield loss, planned in the costing, not a fault.

WORKED EXAMPLE 2 — SHRINKAGE AND RESIDUAL SHRINKAGE TESTS

Shrinkage % = (original length − washed length) / original length × 100.
Unshrunk fabric: 50.0 cm specimen → 46.5 cm after washing: (50.0 − 46.5)/50.0 × 100 = **7.0% shrinkage**.
Sanforized fabric: 50.0 cm → 49.6 cm: (50.0 − 49.6)/50.0 × 100 = **0.8% residual shrinkage** — within the typical ≤1% sanforized claim.
Same formula, two meanings: the first measures what the untreated cloth loses; the second

> measures what the *finished* cloth still has left to lose, and it is the second figure the buyer specifies.

5. Raising, shearing, brushing

- **Raising (napping):** teasing fibre ends out of the yarn surface with wire-covered rollers (card clothing) or teasels to form a soft pile — flannelette, fleece, blankets. Repeated passes raise more pile but weaken the fabric as fibres are pulled from the yarn body; raising is done dry or slightly damp, and often followed by shearing to level the pile.
- **Shearing / cropping:** cutting the surface fibres to a uniform height with a spiral-blade cylinder against a ledger blade — the textile equivalent of a lawnmower. It levels raised pile, cleans loose fibres from smooth fabrics and is essential before printing-quality surfaces and after raising.
- **Brushing:** softer rotating brushes (bristle or nylon) lift a light nap, remove dust and loose fibre, and lay the pile in one direction; used on woollens, velvets and before inspection. Brushing changes feel and appearance without the fibre loss of raising.

6. Singeing as a finishing step

Singeing — passing fabric rapidly over gas flames or heated plates to burn off protruding fibre ends — belongs mainly to pretreatment, but it is also used as a **finishing correction**: on fabrics that have become hairy through raising-adjacent handling, on polyester blends where pilling starts from surface fuzz, and to sharpen the surface before a final calender or a late print/quality upgrade. The principle is differential burning: loose ends, surrounded by air, ignite instantly, while the tight yarn body, moving fast and acting as a heat sink, is unharmed. Speed, flame distance and immediate quenching are the safety controls; over-singeing yellows cotton and melts synthetic tips into hard beads that worsen pilling and dye speck.

7. Heat setting of synthetics

Thermoplastic fibres (polyester, nylon, acrylic, elastane blends) remember the shape in which they were last held hot. **Heat setting** exploits this: fabric held at a controlled width and length on a stenter at high temperature (polyester typically 180–200 °C for 15–30 s) has its internal polymer stresses relaxed and new intermolecular order formed; on cooling, those dimensions become the fabric's stable state. Consequences: wash shrinkage nearly eliminated, creases pressed in stay in, creases acquired in wear relax out, and subsequent processes must stay below the setting temperature or the set is overwritten. Setting *before* dyeing gives level dyeing and stable width in the dye bath; setting after dyeing risks sublimation of disperse dye and shade change — the route is chosen per quality (Topic 1, Section 9).

8. Compacting and emerising

- **Compacting** is the knitwear equivalent of sanforizing: a felt-blanket or rubber-belt compactor overfeeds the knitted fabric and compresses it lengthwise, controlling residual shrinkage, spirality tendency and GSM. Tubular and open-width compactors are standard final machines for cotton knits.
- **Emerising / peach finish (sueding):** the fabric passes over rollers covered with fine emery or silicon-carbide abrasive (or against carbon-brush rollers). The micro-abrasion cuts and fibrillates surface fibres into a dense, very short, uniform fuzz — the soft "peach skin" or suede hand of microfibre polyester and of quality cotton. Control variables are abrasive grit, roller pressure, number of passes and fabric tension; excess emerising thins the fabric and drops tensile strength.

9. Stentering basics

The stenter is the central mechanical finishing machine (drawn in full in Topic 4): fabric is held by its selvedges on pins or clips carried by two diverging chains, overfed slightly to allow lengthwise relaxation, dried and set to a commanded finished width as it travels through hot-air chambers. Stentering delivers the final width, corrects bow and skew (with weft straighteners at entry), dries padded chemical finishes and performs heat setting — mechanical finishing and chemical finishing meet on this one machine, which is why its settings appear in almost every finishing recipe.

10. Comparison — durability of mechanical effects

FINISH	LOCKED BY	SURVIVES WASHING?
Calender lustre (cotton)	Yarn flattening only	No — largely lost in first wash
Embossing on polyester	Heat-set thermoplastic deformation	Yes, if wash temperature stays below set temperature
Raising / emerising	Physical change of surface fibres	Yes — fibres are actually lifted/cut; effect is structural
Sanforizing / compacting	Pre-relaxed, crimp-increased geometry	Yes — residual shrinkage stays low for garment life
Heat setting	Polymer stress relaxation and re-ordering	Yes, up to the setting temperature
Decatising set (wool)	Steam-set keratin under tension	Largely — renewed by pressing in garment care

11. Glossary — key terms

- **Nip:** the contact line between two rollers where pressure is applied to the fabric.
- **Bowl:** a calender roller — heated steel, or resilient cotton/polymer composition.
- **Overfeed:** feeding fabric faster than the chain/machine speed so it can relax and shrink lengthwise under control.
- **Residual shrinkage:** shrinkage remaining in finished fabric, measured by standard wash test.
- **Heat setting:** thermal stabilisation of thermoplastic fabric in a held shape, on cooling made permanent up to the set temperature.
- **Emerising:** fine abrasion of the surface to a short, dense peach-skin fuzz.

12. Key points

- Five variables run every mechanical finish: temperature, moisture, pressure, tension/overfeed and time.
- Sanforizing compresses with a contracting rubber belt; prove it by wash test — 0.8% residual passes a ≤1% claim, 7.0% untreated does not.
- Calender family: swizzing smooths, chasing textures, friction glazes, embossing patterns, Schreiner gives silk-like lustre.
- Structural changes (raising, shrinkage control, heat setting) are durable; surface flattening alone is not.

13. Summary

Mechanical finishing is controlled physics. Decatising steam-sets wool; calenders flatten and pattern surfaces in five distinct modes; the rubber-belt sanforizer buys dimensional peace of mind at the price of planned length loss (50 m in, 47.0 m out at 6%); raising, shearing and brushing engineer the fibre surface; heat setting writes a permanent memory into synthetics; compacting does the same stabilising work for knits; and the stenter holds width and dryness for nearly all of it. Topic 3 adds the chemical half of the finishing department.

14. Practice

1. A 120 m fabric lot is sanforized at 5% compressive shrinkage. What length is delivered? How much extra grey length is needed to deliver exactly 200 m finished?
2. A specimen measures 50.0 cm before washing and 47.8 cm after. Calculate the shrinkage and state whether it could be sold as sanforized (≤1% residual).

3. Choose the calender type for: (a) a glazed chintz curtain, (b) a silk-like cotton blouse fabric, (c) a raised pattern on polyester upholstery.
4. Why is heat setting of polyester done at a fixed width on a stenter rather than loose in an oven?
5. State two risks of over-emerising a lightweight cotton fabric.

Answer key

1. Delivered = 120 × (1 − 0.05) = 114.0 m. For 200 m finished: grey needed = 200 / 0.95 = 210.5 m (about 10.5 m extra).
2. Shrinkage = (50.0 − 47.8)/50.0 × 100 = 4.4%. No — far above the ≤1% residual limit; the fabric needs compressive shrinkage treatment (or more of it).
3. (a) Friction calender. (b) Schreiner calender. (c) Embossing calender (durable because the pattern is heat-set into the thermoplastic).
4. Setting fixes the dimensions in which the fabric is held. Loose fabric would relax unevenly — width, length and skew uncontrolled — and would set in creases; the stenter's pins/clips and overfeed hold exactly the target geometry while the polymer relaxes.
5. Loss of fabric substance/weight and a measurable drop in tensile and tear strength; also uneven surface and increased pilling if fibres are over-cut (any two).

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib